



The increasing number of pathogens that can resist current antibiotics has increased until a point that the World Health Organization has declared it a major threat to global health. Consequently, much research has been focused on finding new therapies that exhibit novel mechanisms of action. In this regard, as demonstrated in the pioneer work of R. Ghadiri in 1993, who demonstrated the use of cyclic peptides that could self-assemble into supramolecular nanotubes preferentially in lipidic membranes. This fact provides them a huge potential as antimicrobial and/or antineoplastic agents. The culmination of the project would be the creation of the first computational platform that allows an automated design and optimization of antimicrobial and antineoplastic peptidomimetics based on CPs.